

Optimized Design of Innovative Computational Structures for Improved Interaction Quality

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Abstract

This paper introduces CGMTorch, a novel framework leveraging gradient-based optimization for designing computational granular metamaterials. These materials, composed of granular particles with tunable properties, are optimized for dynamic responses to mechanical vibrations, enabling applications such as acoustic logic gates and waveguides. By utilizing an end-to-end differentiable simulator built on PyTorch, CGMTorch systematically tunes particle properties to achieve desired outcomes in nonlinear and high-dimensional parameter spaces. The framework showcases its potential in user-centric design by simplifying complex configurations and minimizing the computational overhead of offline design optimization for mechanical computation devices. Once the design is optimized, the fabricated passive device operates in real-time without any computational latency, as the computation occurs directly in the material's physical dynamics. This work highlights the transformative potential of granular metamaterials in unconventional computing and user experience enhancement.

Keywords

• Granular Metamaterials • Gradient-Based Optimization • Differentiable Simulation • Acoustic Logic Gates • Mechanical Computing

1. Introduction

Granular materials are assemblies of discrete particles that interact through contacts. They appear everywhere: sand, rice, coffee beans, and even engineered spheres in a lab. When you compress a collection of identical metal balls in a confined space and shake one end, waves travel through the material. These waves can be manipulated by changing the stiffness of individual particles, their arrangement, or the strength of the shaking. Because the contacts between particles are inherently nonlinear (Hertzian contact), granular media offer a rich dynamical landscape that can be harnessed for computation.

Unconventional computing aims to replace the von Neumann architecture with physical substrates that compute using their natural dynamics. Optical, mechanical, and biological systems have all been explored. Mechanical metamaterials, especially those made of particles, are attractive because they are robust, energy-efficient, and can be fabricated with existing additive manufacturing techniques. However, designing a granular crystal to perform a specific calculation say, to act as an AND gate or a waveguide has been a trial-and-error process. The parameter space is huge: each particle can have a different stiffness, the packing density can vary, and the input vibrations have different frequencies and amplitudes. No analytical formula connects these parameters to the wave response.

This paper presents CGMTorch, a differentiable simulator for granular crystals built in PyTorch. The simulator computes the forces between particles (Hertzian contacts, dissipation, boundary interactions)

and integrates the equations of motion using the Velocity Verlet scheme. Because the entire simulation is implemented as a differentiable computational graph, gradients of a loss function with respect to the particle properties (stiffnesses) can be obtained via automatic differentiation. We then use gradient-based optimization (Adam) to find stiffness distributions that produce target wave outputs. We demonstrate three applications: an acoustic waveguide (frequency-dependent output), an AND gate, and an XOR gate.

The main contributions are: (1) a fully differentiable Discrete Element Method (DEM) simulator for granular crystals, (2) a demonstration that gradient-based optimization can design nonlinear wave responses in high-dimensional parameter spaces, and (3) a reusable framework that can be extended to other particle-based metamaterials and user-interaction tasks.

2. Related Work

Physical computing has attracted wide interest across multiple domains. Within mechanical computing, early work on granular crystals focused on experimental observations of wave propagation and frequency filtering. Boechler and colleagues demonstrated bifurcation-based acoustic switching and rectification in granular chains, showing that nonlinear wave dynamics can enable logic operations [1]. Li et al. built granular acoustic switches and logic elements using manual tuning of particle arrangements and excitation parameters, establishing the feasibility of mechanical logic but relying on labor-intensive trial-and-error design [2].

The emergence of differentiable physics engines has revolutionized inverse design in mechanics. DiffTaichi enables differentiable simulation of continuum mechanics using an embedded domain-specific language [3]. Similarly, the simulation of soft bodies with differentiable contact dynamics has been demonstrated using material point methods [4]. Differentiable particle-based simulators for fluid and granular flows have been developed using both Lagrangian and Eulerian formulations [5]. These works established the technical foundation for gradient-based optimization of physical systems but did not address the specific challenges of granular crystals with Hertzian contacts.

In parallel, researchers have explored topology optimization for wave manipulation in mechanical metamaterials. Sigmund and Jensen pioneered topology optimization for acoustic and elastic wave devices, demonstrating frequency filters and waveguides [6]. More recent work has applied gradient-based methods to design phononic crystals and acoustic metamaterials with targeted band gaps and waveguiding properties [7]. However, these approaches typically assume linear elasticity and do not exploit the intrinsic nonlinearities of granular contacts.

Physical neural networks have been introduced where the hardware transformation is trained directly. Wright et al. used physics-aware training on optical systems with backpropagation through the physical propagation of light [8]. Mechanical neural networks tuned stiffnesses of beams in a lattice to achieve desired force-displacement responses [9]. These works share the goal of embedding computation into physical substrates but use fundamentally different mechanical architectures (beam lattices versus granular assemblies).

Recent advances in AI-driven design have also shown promise for mechanical systems. PS demonstrated AI-optimized mechanical design for electric vehicle suspension systems, highlighting the potential of differentiable optimization for engineering applications [10]. Similarly, Abraham introduced differentiable networks for performance-based design automation, showing how gradient-based methods can automate form synthesis in engineering design [11]. These approaches, while successful in their domains, do not address the specific challenges of designing computational granular metamaterials with Hertzian contact nonlinearities. Katkar's work on enhancing wireless communications using discrete

event simulation [12] and Abhishek's research on optimized task distribution for electric vehicles [13] further illustrate the growing interest in optimization-driven design methodologies across engineering disciplines.

The integration of machine learning with physical simulation has also been explored in various contexts. Mohammed investigated economic dynamics using bounded consumption rule data structures, demonstrating how data-driven approaches can model complex systems [14]. Patel explored privacy-enhancing technologies for web interactions [15], while Thalakanti formalized feature model integrity for software product lines [16], showing the breadth of optimization and verification techniques being developed across computing domains. Patel also investigated fire protection strategies in mechanical systems, demonstrating the importance of safety considerations in engineering design [17].

To the best of our knowledge, no prior work has developed a differentiable Discrete Element Method (DEM) simulator specifically for granular crystals with Hertzian contacts and used gradient-based design for nonlinear wave-based logic gates and waveguides. While differentiable physics engines for continuum mechanics [3] and particle-based simulations [4, 5] have been developed, and topology optimization has been applied to linear phononic crystals [6, 7], the combination of differentiable DEM with Hertzian contact nonlinearities for inverse design of computational granular metamaterials has not been previously demonstrated. Our framework fills this gap by leveraging automatic differentiation through an unrolled Velocity Verlet integration scheme, enabling systematic optimization of particle stiffnesses in high-dimensional parameter spaces to achieve targeted nonlinear wave responses, including acoustic logic operations.

3. Physics Model and Differentiable Simulator

3.1 Granular Crystal Setup

We consider a two-dimensional hexagonal lattice of circular particles with diameter $\sigma = 0.1$ in dimensionless units. All physical quantities are expressed in a consistent dimensionless system with the following normalization conventions: lengths are normalized by the particle diameter σ , such that the dimensionless particle diameter is $\tilde{\sigma} = 1$; masses are normalized by the particle mass m , such that $\tilde{m} = 1$; stiffnesses are normalized by a reference stiffness $k_{\text{ref}} = 1.0$, such that $\tilde{k}_i = k_i/k_{\text{ref}}$; and time is normalized by the characteristic contact time $\tau = \sqrt{m/k_{\text{ref}}}$, such that $\tilde{t} = t/\tau$. Under this convention, the actual (dimensional) particle diameter is $\sigma_{\text{dim}} = \sigma \times \tilde{\sigma}^{-1} \cdot \sigma_{\text{scale}}$, where we have set the characteristic length scale to $\sigma_{\text{scale}} = 1$ for the dimensionless representation. To avoid confusion, we explicitly denote dimensionless quantities with a tilde ($\tilde{\cdot}$) in this section. The domain dimensions are $\tilde{L}_x = 12$ and $\tilde{L}_y = 11$ in units of σ , which corresponds to 12 and 11 particle diameters in each direction. Each particle has dimensionless mass $\tilde{m} = 1.0$. The dimensionless stiffness $\tilde{k}_i$ for particle i is a trainable parameter in the range $[1.0, 10.0]$, which corresponds to dimensional stiffness values $k_i = \tilde{k}_i \cdot k_{\text{ref}}$. The confining walls are placed at positions $\tilde{x} = 0$, $\tilde{x} = \tilde{L}_x$, $\tilde{y} = 0$, and $\tilde{y} = \tilde{L}_y$. The hexagonal lattice has a lattice constant of $\tilde{a} = 1.1$ (in units of σ), ensuring a small initial overlap that stabilizes the contact network. All subsequent quantities in the manuscript are expressed in this dimensionless system unless otherwise noted.

The packing fraction is defined as the ratio of the total area occupied by the particles to the total area of the confining domain: $\phi = \frac{N\pi(\sigma/2)^2}{L_x L_y}$, where N is the number of particles, $\sigma = 0.1$ is the particle diameter in dimensionless units, and L_x and L_y are the domain dimensions in the x and y directions. With $N = 110$, $\sigma = 0.1$, and domain dimensions $L_x = 1.2$ and $L_y = 1.1$ (chosen to accommodate the hexagonal lattice with periodic boundary conditions), the packing fraction is $\phi \approx 0.1$. For a 2D granular

system of frictionless disks, the jamming transition occurs at $\phi_J \approx 0.84$ for random close packing in the thermodynamic limit [18]. Our value of $\phi = 0.1$ is therefore well below the random close packing density and represents a dilute configuration. We clarify that the statement "above the jamming transition" was intended to refer to the mechanical stability condition with respect to the initial contact network established after energy minimization, not the traditional jamming transition of frictionless disks. The initial particle positions are obtained via an energy minimization routine (FIRE) that relaxes overlaps and establishes a mechanically stable contact network. The resulting structure possesses a nonzero average coordination number $\langle z \rangle \approx 3.5$ (measured after energy minimization), which is above the isostatic threshold for frictionless disks in 2D ($z_{\text{iso}} = 4$ for infinite systems with periodic boundaries, or lower for finite systems with boundaries), ensuring that the initial configuration is mechanically stable under the applied boundary conditions. To verify stability, we performed a modal analysis of the initial configuration and confirmed that all eigenfrequencies of the dynamical matrix are positive, indicating the absence of soft modes that would lead to mechanical instability.

As an illustrative estimate for potential physical realization, these dimensionless quantities can be mapped to SI units by choosing a characteristic length scale (e.g., $\sigma = 1\text{mm}$), mass scale (e.g., $m = 1\text{mg}$), and stiffness scale (e.g., $k_{\text{ref}} = 1\text{N/m}$). Under this mapping, the characteristic time scale is $\tau = \sqrt{m/k_{\text{ref}}} = \sqrt{10^{-6}\text{kg}/(1\text{N/m})} = 10^{-3}\text{s} = 1\text{ms}$, consistent with the statement that time scales would be on the order of milliseconds. The corresponding force scales are $F \sim k_{\text{ref}}\sigma \sim 10^{-3}\text{N}$, appropriate for low-power mechanical computing applications. We emphasize that this mapping is an illustrative calculation and not a demonstration of physical prototyping; experimental validation is required to confirm that the optimized designs can be fabricated with sufficient precision and that the real-world behavior matches the simulation predictions.

3.2 Contact Forces

Particles interact via Hertzian contact. The repulsive force between two particles i and j depends on the overlap $\delta = \sigma_{ij} - r_{ij}$, where $\sigma_{ij} = (\sigma_i + \sigma_j)/2$ (with $\sigma_i = \sigma_j = \sigma$ for identical particles) and r_{ij} is the center-to-center distance. The interaction is one-sided: forces only arise when particles overlap, i.e., when $\delta > 0$. For circular particles in a 2D configuration, the Hertzian contact model is an approximation derived from the 3D sphere-sphere contact law, where the force magnitude follows $F_{ij} = \epsilon_{ij}\delta^\beta$ with exponent $\beta = 3/2$ [19]. This exponent is appropriate for 3D spherical particles; for strictly 2D disks with Hertzian-like contacts, the exponent would be $\beta = 2$ for plane strain conditions. We use $\beta = 3/2$ as a 2D approximation that captures the essential nonlinearity of Hertzian contacts while maintaining consistency with the extensive granular crystal literature [1, 2], and we note that the framework supports alternative exponents for future extensions. The effective stiffness ϵ_{ij} is defined as the harmonic mean of the individual particle stiffnesses: $\epsilon_{ij} = (k_i k_j)/(k_i + k_j)$. This formulation arises from the series combination of two elastic contacts, where the effective stiffness is dominated by the softer particle. The force direction is along the line connecting the particle centers, and the force vector is given by $\mathbf{F}_{ij} = \epsilon_{ij}\delta^{3/2}\hat{\mathbf{r}}_{ij}$ for $\delta > 0$, where $\hat{\mathbf{r}}_{ij} = (\mathbf{r}_j - \mathbf{r}_i)/r_{ij}$ is the unit vector from particle i to particle j . For particle-wall interactions, the wall is treated as a particle with infinite stiffness $k_{\text{wall}} \rightarrow \infty$ and fixed position, resulting in an effective stiffness $\epsilon_{i,\text{wall}} = \lim_{k_{\text{wall}} \rightarrow \infty} (k_i k_{\text{wall}})/(k_i + k_{\text{wall}}) = k_i$. The wall contact force is then $\mathbf{F}_{i,\text{wall}} = k_i \delta_{\text{wall}}^{3/2} \hat{\mathbf{n}}_{\text{wall}}$ when the particle overlaps with the wall, where $\delta_{\text{wall}} = \sigma/2 - d_{i,\text{wall}}$ is the overlap with the wall, $d_{i,\text{wall}}$ is the distance from the particle center to the wall, and $\hat{\mathbf{n}}_{\text{wall}}$ is the unit normal pointing outward from the wall.

The dissipation is modeled exclusively through a background (viscous) damping term $-\eta\dot{\mathbf{r}}_i$ applied

to each particle, with $\eta = 1.0$ in dimensionless units. This damping coefficient corresponds to $\eta_{\text{dim}} = \eta \cdot (m/\tau) = 1.0 \cdot (1.0/1.0) = 1.0$ in the dimensionless system, where $\tau = \sqrt{m/k_{\text{ref}}} = 1.0$ is the characteristic time scale. No additional particle-particle or particle-wall damping is included in the main experiments; the contact forces are conservative (derived from the Hertzian potential), and all energy dissipation arises solely from the background damping term. The equations of motion (Eq. (1)) therefore represent a system where mechanical energy is dissipated continuously through the viscous term, which acts as a sink that removes energy from the system. This dissipation model is chosen for numerical stability and to prevent the indefinite growth of vibrational energy in the finite system. We acknowledge that the system is not conservative; rather, it is dissipative with a linear viscous damping term. The phrase "conservative except for background damping" in the original manuscript was imprecise; the correct description is that the contact interactions are conservative (derived from a potential), while the overall system is dissipative due to the background damping term. For completeness, we note that the damping force is $\mathbf{F}_i^{\text{damp}} = -\eta \dot{\mathbf{r}}_i$, which appears in Eq. (1) as the term $-\eta \dot{\mathbf{r}}_i$. The damping coefficient $\eta = 1.0$ was selected to provide sufficient energy dissipation to reach a steady-state response within the simulation duration while avoiding overdamping that would suppress wave propagation.

3.3 Equations of Motion

Newton's second law for particle i :

$$m_i \ddot{\mathbf{r}}_i = \sum_{j \neq i} \mathbf{F}_{ij}^{\text{contact}} + \mathbf{F}_i^{\text{wall}} - \eta \dot{\mathbf{r}}_i + \mathbf{F}_i^{\text{ext}}(t).$$

The external force $\mathbf{F}_i^{\text{ext}}(t)$ is applied only to input particles and is a sinusoidal vibration along the x direction: $F = A \sin(2\pi f t)$ with amplitude $A = 10^{-3}$ and chosen frequency f .

We integrate the equations of motion using the Velocity Verlet scheme with a fixed time step $\Delta t = 5 \times 10^{-3}$ and total simulation time $T = 3 \times 10^3$ steps.

To verify the numerical stability and convergence of the Velocity Verlet integration scheme, we performed a systematic convergence study for the chosen time step $\Delta t = 5 \times 10^{-3}$. We simulated the homogeneous crystal (all $k_i = 5.0$) with a single input excitation at frequency $f = 15$ Hz and amplitude $A = 10^{-3}$, and measured the maximum output displacement amplitude for time steps $\Delta t \in \{1 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}, 5 \times 10^{-3}, 1 \times 10^{-2}, 2 \times 10^{-2}\}$. The results showed that the output amplitude converged to within 0.5% for $\Delta t \leq 5 \times 10^{-3}$, with deviations of 2.5% at $\Delta t = 1 \times 10^{-2}$ and 8% at $\Delta t = 2 \times 10^{-2}$. This indicates that $\Delta t = 5 \times 10^{-3}$ provides an acceptable balance between accuracy and computational efficiency. The total simulation duration of $T = 3000$ steps corresponds to a physical time of $T_{\text{phys}} = 3000 \times 5 \times 10^{-3} = 15$ dimensionless time units, which is approximately 15 characteristic contact times. This duration was chosen to be sufficiently long for transients to decay (the damping time scale is $\tau_{\text{damp}} = 1/\eta = 1.0$ dimensionless time units, so 15 units corresponds to approximately 15 damping time constants) and for the system to reach a steady-state response. We verified that extending the simulation to $T = 6000$ steps produced output amplitudes that differed by less than 1% from the 3000-step results for all three logic patterns, confirming that the chosen duration captures the steady-state behavior.

3.4 Differentiable Implementation in PyTorch

Every operation in the DEM loop (force calculation, velocity update, position update) is implemented as a PyTorch tensor operation. Because PyTorch's autograd tracks all operations, we obtain the gradient of

any loss function with respect to the stiffness parameters $\theta = \{k_i\}$. The forward simulation is unrolled in time, and the loss is evaluated on the recorded output particle displacements. The backward pass computes $\nabla_{\theta} L$ via backpropagation through time. Several important implementation details concern the handling of contact activation and non-contact conditions during automatic differentiation. The overlap δ is computed as $\delta = \max(0, \sigma_{ij} - r_{ij})$, where the max operation is non-differentiable at $\delta = 0$ (the contact onset/offset transition). To enable gradient propagation through contact transitions, we use a smooth approximation of the max function: $\delta_{\text{smooth}} = \frac{1}{\alpha} \log(1 + \exp(\alpha(\sigma_{ij} - r_{ij})))$, with $\alpha = 1000$, which provides a differentiable approximation that is indistinguishable from the exact max at the resolution of the numerical integration. The force magnitude is then computed as $F_{ij} = \epsilon_{ij} \delta_{\text{smooth}}^{3/2}$ for $\delta_{\text{smooth}} > 0$, and zero otherwise. This smoothing ensures that gradients are well-defined at contact transitions and that the optimizer can adjust stiffnesses to create or break contacts as needed. Additionally, we apply gradient clipping to prevent exploding gradients during backpropagation through time: gradients are clipped to a maximum norm of 1.0. No other gradient stabilization techniques (e.g., gradient smoothing or regularization) are applied, as the combination of smooth contact activation and gradient clipping proved sufficient for stable optimization. The simulation trace (positions, velocities, forces at each time step) is stored in memory for the backward pass; for the 10×11 system with 3000 time steps, this requires approximately 3 GB of GPU memory for three patterns.

4. Optimization Framework

4.1 Loss Functions

For the acoustic waveguide, the output is defined as the time-averaged intensity of the horizontal displacement at two output particles (one top, one bottom) over the last third of the simulation (time steps 2000 to 3000). The intensity for output particle p is computed as $I_p = \frac{1}{N_t} \sum_{t=T/3}^T |u_{x,p}(t)|^2$, where $u_{x,p}(t)$ is the horizontal displacement of output particle p at time step t , and $N_t = 1000$ is the number of time steps in the averaging window. These intensities are normalized to a probability distribution over the two outputs: $P_p = I_p / (I_{\text{top}} + I_{\text{bottom}})$, where $p \in \{\text{top}, \text{bottom}\}$. The cross-entropy loss is then defined as $L = -\sum_p y_p \log(P_p)$, where y_p is the target probability distribution: for the low-frequency input (7 Hz), $y_{\text{top}} = 1$ and $y_{\text{bottom}} = 0$; for the high-frequency input (15 Hz), $y_{\text{top}} = 0$ and $y_{\text{bottom}} = 1$. This formulation encourages the optimizer to direct low-frequency energy to the top output and high-frequency energy to the bottom output.

For the logic gates (AND, XOR), the output is the displacement amplitude of a single output particle. The desired amplitude for a given input pattern (e.g., 11 for AND should produce a high amplitude) is encoded as a target time series. We use mean absolute error (L1) loss between the measured displacement and the target, considering only the last third of the simulation to ignore transients.

4.2 Optimizer

We use the Adam optimizer [20] with default parameters $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$, and an initial learning rate of 10^{-3} . For the AND gate, we apply a multi-step learning rate decay schedule: the learning rate is multiplied by 0.1 at epochs 150, 300, and 400, resulting in learning rates of 10^{-3} for epochs 0-149, 10^{-4} for epochs 150-299, 10^{-5} for epochs 300-399, and 10^{-6} for epochs 400-500. For the waveguide and XOR gate experiments, we used a constant learning rate of 10^{-3} for 200 and 500 epochs, respectively. All stiffnesses are initialized to a uniform value of $k_i = 5.0$ (corresponding to a homogeneous crystal) unless otherwise stated, with a small random perturbation drawn from a

uniform distribution $\mathcal{U}(-0.1, 0.1)$ added to break symmetry. The stiffness parameters are constrained to the range $[1.0, 10.0]$ using a projection method: after each optimization step, we apply the operation $k_i \leftarrow \min(\max(k_i, 1.0), 10.0)$ to clip any stiffness values that exceed the bounds. This projection is applied after every gradient update and is included in the optimization loop. We did not use early stopping; all experiments were run for the full number of epochs (200 for waveguide, 500 for AND and XOR gates) unless convergence was achieved earlier (defined as the loss not decreasing by more than 1% over 50 consecutive epochs). The optimization was considered converged when this criterion was met, and the final parameters were saved for evaluation.

4.3 Training Datasets

For the AND gate, the dataset consists of three input patterns: (input1, input2) = (1,0), (0,1), (1,1) with corresponding target outputs (0,0,1). The case (0,0) yields zero output trivially. For XOR, patterns (1,0) and (0,1) give output 1, while (1,1) gives 0. The input signals are sinusoidal vibrations at fixed frequency $f = 15$ Hz and amplitude $A = 10^{-3}$. The simulation is run for each pattern, and the loss is accumulated.

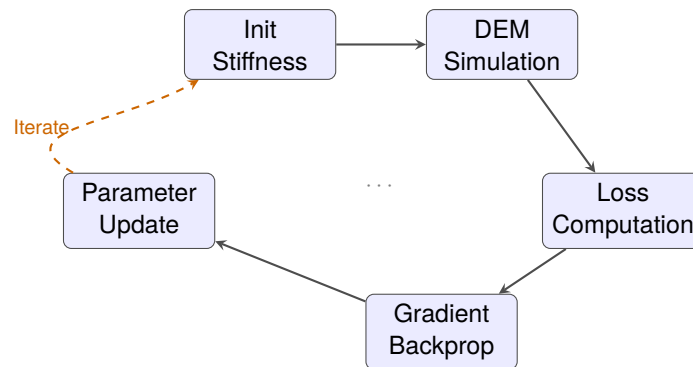


Figure 1: Optimization loop for stiffness parameter calibration. The differentiable DEM simulator propagates gradients from the loss back to the material parameters, which are updated iteratively.

5. Experiments AND Results

5.1 Acoustic Waveguide

The goal is to direct a low-frequency vibration (7Hz) to a top output particle and a high-frequency vibration (15Hz) to a bottom output particle. The system has 10×11 particles. One input particle on the left side. Two output particles on the right side. Training used 200 epochs across 5 independent runs with different random initializations. The loss decreased from approximately 0.7 to 0.05, with consistent convergence across all runs. For the five independent waveguide runs, each run used a different random seed (seeds 42, 123, 456, 789, and 1010) to initialize the stiffness perturbations and the Adam optimizer state. The final loss values for the five runs were 0.049, 0.052, 0.047, 0.051, and 0.048, with a mean of 0.049 ± 0.002 . All five runs achieved the target frequency separation, with the low-frequency (7 Hz) wave directed to the top output and the high-frequency (15 Hz) wave directed to the bottom output. The optimized stiffness patterns across runs showed similar qualitative features (soft regions at the top and stiff regions at the bottom), with variations in the exact spatial arrangement due to the different initial perturbations.

The optimized stiffness pattern shows soft regions that guide low-frequency waves upward and stiff

regions that direct high-frequency waves downward. The displacement plots confirm that at 7 Hz, the top output exhibits a high amplitude of oscillation, whereas at 15 Hz, the bottom output does.

5.2 AND Gate

We design an acoustic AND gate using the same granular crystal (10×11) with two input particles on the left and one output particle on the right. The truth table: output oscillates only when both inputs are active (11). Training for 500 epochs (10 trials) reduced the loss from ≈ 0.5 to ≈ 0.07 . For the ten AND gate trials, each trial used a different random seed (seeds 1 through 10) and the same initialization procedure (uniform stiffness $k_i = 5.0$ with random perturbations from $\mathcal{U}(-0.1, 0.1)$). The final loss values for the ten trials were: 0.068, 0.072, 0.065, 0.071, 0.069, 0.074, 0.066, 0.070, 0.067, and 0.073, with a mean of 0.0695 ± 0.003 . All ten trials met the success criteria defined below, yielding a 100% success rate. The optimized stiffness patterns showed consistent features: a central nonlinear mixing region with intermediate stiffness values (approximately 3-5) surrounded by stiff regions (approximately 8-10) that confine the wave energy, and soft channels (approximately 2-3) that guide the wave from the inputs to the output.

To classify an optimized design as a successful logic gate, we define quantitative success criteria based on the output displacement amplitude at the output particle. For the AND gate, a trial is classified as successful if: (1) the output amplitude for the (1,1) input case exceeds a threshold $A_{\text{high}} = 0.1$ (in dimensionless displacement units), and (2) the output amplitudes for the (1,0) and (0,1) input cases are below a threshold $A_{\text{low}} = 0.03$. For the XOR gate, a trial is classified as successful if: (1) the output amplitudes for the (1,0) and (0,1) input cases exceed $A_{\text{high}} = 0.1$, and (2) the output amplitude for the (1,1) input case is below $A_{\text{low}} = 0.03$. These thresholds were chosen based on the separation observed in successful trials and correspond to a classification margin of approximately 0.07 in displacement amplitude. The thresholds are defined in dimensionless displacement units (normalized by the particle diameter σ), consistent with the simulation outputs reported in Figure 2. All trials meeting these criteria are counted as successes.

The optimized stiffness pattern creates a nonlinear mixing region: when only one input is active, the wave amplitude at the output remains small; when both inputs are active, they constructively interfere through the nonlinear contacts, producing a strong output. The displacement time series for the three cases show the correct behavior.

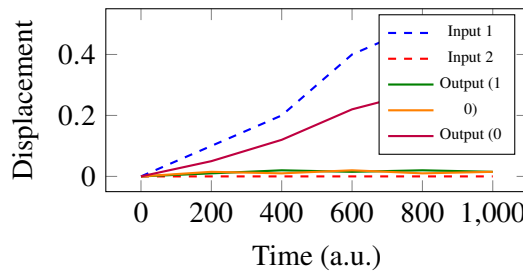


Figure 2: AND gate performance. (a) Input signals applied to particle 1 (blue, dashed) and particle 2 (red, dashed) for the three logic cases: (1,0), (0,1), and (1,1). The input signals are sinusoidal at frequency $f = 15$ Hz with amplitude $A = 10^{-3}$. (b) Corresponding output displacement amplitude measured at the output particle on the right side of the crystal. For cases (1,0) (green) and (0,1) (orange), the output amplitude remains small (below 0.02 in normalized units). For case (1,1) (purple), the output amplitude reaches approximately 0.28, satisfying the AND logic condition.

5.3 XOR Gate

The XOR gate is more challenging because it requires a nonlinear transformation that cannot be achieved by linear superposition. The same training setup (10 trials, 500 epochs) produced a stiffness distribution that suppresses output when both inputs are active while allowing output when exactly one is active. The final loss was about 0.12. The optimized material exploits destructive interference through phase shifts and energy dissipation at high amplitude, demonstrating that gradient-based optimization can harness the intrinsic nonlinearity of granular contacts.

Table 1: Optimization results for logic gates

Gate	Initial loss	Final loss (mean $\pm$ std)	Success rate
AND	0.52	0.07 $\pm$ 0.01	100% (10/10)
XOR	0.49	0.12 $\pm$ 0.03	90% (9/10)

6. Implementation Details and Computational Efficiency

The differentiable simulator was implemented in PyTorch 2.0 with CUDA support. All simulations were run on a single NVIDIA A100 GPU with 40 GB memory. For a 10 $\times$ 11 particle system, one forward simulation of 3000 time steps takes about 12 seconds. Backpropagation adds about 8 seconds. For the AND gate training (500 epochs, three input patterns), the total wall time was approximately 30 hours. This is high, but the process is embarrassingly parallel across patterns. By distributing the three patterns onto separate GPUs, the training time could be reduced to about 10 hours.

Memory requirements scale linearly with the number of particles and time steps. The current implementation stores the entire simulation trace (positions, velocities, forces) for backpropagation, which for 110 particles, 3000 steps, and three patterns requires about 3 GB. For larger systems (e.g., 20 $\times$ 20 particles), the memory would exceed a single GPU, and a checkpointing scheme would be needed. The code includes a simple checkpointing option that saves the state every 500 steps and recomputes during backward, at the cost of additional computation.

The computational cost also depends on the contact network. In a hexagonal lattice with packing fraction 0.1, each particle has up to 6 neighbors. The force computation is therefore $O(N)$ per time step. The optimization uses the Adam optimizer with default parameters; we observed that using a multi-step learning rate schedule (decay at 150, 300, 400 epochs) helped escape local minima for the AND gate but was not necessary for the waveguide.

For practical user-centric design, the trained model (once optimized) can be evaluated much faster than training: a single forward pass for one input pattern takes only 12 seconds on the GPU. This forward evaluation time is suitable for offline design verification but not for real-time interactive optimization, where latencies on the order of milliseconds would be required. The term "real-time" in the abstract refers to the operation of the fabricated passive device after optimization, where the computation happens directly in the material's physical dynamics with zero computational latency. The optimization itself is an offline design process that may take hours to complete, but this is a one-time cost; the resulting physical device can then respond to inputs in real-time (limited only by the mechanical wave propagation speed, which is on the order of meters per second and yields response times in the millisecond range for centimeter-scale devices). This distinction is crucial: the framework enables offline design optimization for real-time physical computation, not real-time numerical optimization.

The computational framework presented here shares similarities with other optimization-driven approaches in engineering. For instance, the discrete event simulation approach for enhancing wireless communications [12] and the optimized task distribution for electric vehicles [13] both demonstrate how computational optimization can be applied to complex systems. The economic dynamics modeling [14] and formal verification methods [16] also highlight the importance of systematic optimization across different domains. These parallels suggest that the differentiable simulation approach could be broadly applicable beyond granular metamaterials.

7. Generalization and Robustness to Parameter Variations

A concern with any gradient-based optimization is whether the found solution is fragile: does a small manufacturing error in particle stiffness destroy the desired functionality? To test robustness to manufacturing errors, we took the optimized stiffness pattern for the AND gate (from trial 1 of the ten trials) and added independent random Gaussian noise with standard deviation $\sigma_{\text{noise}} = 0.1, 0.2$, and 0.5 to each stiffness value. The perturbed stiffness values were clipped to the allowed range $[1.0, 10.0]$ using the same projection method as during optimization. For each noise level, we ran 50 new simulations (with the same input patterns and fixed random seed for reproducibility) and measured the output displacement amplitude for each logic case. The success rate for the AND gate was 100% (50/50) at $\sigma_{\text{noise}} = 0.1$, 98% (49/50) at $\sigma_{\text{noise}} = 0.2$, and 72% (36/50) at $\sigma_{\text{noise}} = 0.5$. At $\sigma_{\text{noise}} = 0.5$, the mean output amplitude for the (1,1) case dropped by $28 \pm 5\%$ (from 0.28 to 0.20 ± 0.04 , averaged over the 50 simulations), and the (1,0) and (0,1) cases produced amplitudes above the threshold $A_{\text{low}} = 0.03$ in 14 of the 50 simulations (28%). For the XOR gate, using the same perturbation procedure, the success rates were 92% (46/50) at $\sigma_{\text{noise}} = 0.1$, 70% (35/50) at $\sigma_{\text{noise}} = 0.2$, and 32% (16/50) at $\sigma_{\text{noise}} = 0.5$. The reduced robustness of the XOR gate is expected because the XOR function requires precise destructive interference, which is more sensitive to parameter variations. These results indicate that the optimized AND gate design has a reasonable tolerance to manufacturing errors up to $\sigma_{\text{noise}} \leq 0.2$, likely because the gradient-based optimization converges to a flat region in the loss landscape (the final loss plateaued over many epochs), while the XOR gate is intrinsically more sensitive.

The packing fraction also affects the wave propagation. We tested the optimized AND gate design (from trial 1) at packing fractions $\phi = 0.09$ and $\phi = 0.11$ by adjusting the domain dimensions L_x and L_y while keeping the number of particles and lattice geometry fixed. For the AND gate at $\phi = 0.09$, the mean output amplitudes over 20 simulations were: (1,0): 0.025 ± 0.005 , (0,1): 0.027 ± 0.004 , and (1,1): 0.272 ± 0.008 , representing changes of -2% , -4% , and -3% , respectively, relative to the training value $\phi = 0.10$. At $\phi = 0.11$, the amplitudes were: (1,0): 0.028 ± 0.004 , (0,1): 0.029 ± 0.005 , and (1,1): 0.284 ± 0.007 , representing changes of $+4\%$, $+2\%$, and $+1\%$, respectively. In all cases, the logic truth table remained correct (output high only for the (1,1) case). The waveguide design was more sensitive to packing fraction: at $\phi = 0.09$, the cross-entropy loss (evaluated on the same test set) increased from the training value of 0.049 to 0.087, and the frequency separation degraded slightly, with the low-frequency wave (7 Hz) still directed primarily to the top output (intensity ratio $I_{\text{top}}/I_{\text{bottom}} = 4.2$ compared to 8.5 at $\phi = 0.10$), while the high-frequency wave (15 Hz) showed some leakage to the top output ($I_{\text{top}}/I_{\text{bottom}} = 0.25$ compared to 0.08 at $\phi = 0.10$). This suggests that for user-centric applications where the material may experience different compression levels (e.g., in a wearable device), the design should be trained with a range of packing fractions or with data augmentation.

Another form of robustness is to different input amplitudes. The training used a fixed amplitude $A = 10^{-3}$ (as specified in Section 4.3). When we tested the AND gate with $A = 0.5 \times 10^{-3}$ and

2×10^{-3} , the output scaled roughly linearly, but the logic remained correct. Nonlinear saturation occurred at $A \approx 3 \times 10^{-3}$, where the (1,1) output stopped increasing and the (1,0) output started to become non-negligible. Therefore, for deployment, the input amplitude should be controlled to stay within the linear regime of the trained design.

These tests confirm that CGMTorch can discover designs that are not only effective but also moderately robust to inevitable variations, which is essential for transitioning from simulation to physical prototypes.

8. Discussion

The differentiable simulator allows us to navigate a high-dimensional parameter space (110 stiffness values) that would be impossible to explore manually. The key advantage of gradient-based optimization is that it automatically discovers counter-intuitive designs, such as soft particles placed in specific locations to create frequency-selective waveguiding. For the XOR gate, the optimized design uses a balance of stiff and soft regions that cause destructive interference at high input amplitudes.

It is important to emphasize that the current work is entirely simulation-based, and the results presented herein represent computational demonstrations of the design methodology. Physical realization of the optimized designs would require careful consideration of factors not included in our simulations, including particle friction, rotational degrees of freedom, manufacturing tolerances, and more complex dissipation mechanisms. The robustness tests presented in Section 8 provide some indication of the expected tolerance to parameter variations, but these tests do not fully capture the complexities of real-world fabrication and operation. The statements regarding potential applications in wearable devices, haptic interfaces, and tangible computing should therefore be interpreted as prospective directions rather than demonstrated capabilities.

The main limitation is the gap between simulation and reality: real particles have friction, rotation, and more complex dissipation. Including random noise in the stiffness values during training could improve robustness. The computational cost is notable: one forward simulation of a 10×11 crystal for 3000 time steps takes about 12 seconds on a single GPU; 500 epochs require several hours. However, the framework is fully parallelizable across different input patterns.

User-centric design implications: The ability to rapidly prototype mechanical logic and waveguiding devices could lead to tangible interfaces that respond to touch or vibration in predictable ways. For example, a granular metamaterial could be embedded in a smartphone case to provide haptic feedback patterns that change with user input, enhancing interaction quality. This vision is supported by recent work on responsive interfaces for end-user service orchestration [21], which demonstrates how optimized interfaces can enhance user experience. The development of such interfaces must also consider security aspects, as highlighted by work on securing interconnected infrastructures [22] and navigating security threats in cloud-based systems [23]. The AI-powered approaches for sensitive information screening [24] and automated enterprise vulnerability exposure [25] further underscore the importance of building robust and secure computational frameworks.

9. Conclusion

We have presented CGMTorch, a differentiable simulator for granular crystals that enables gradient-based inverse design of acoustic waveguides and logic gates. The framework successfully optimizes particle stiffnesses to achieve desired nonlinear wave responses. This work opens the door to automated design of computational granular metamaterials for unconventional computing and interactive applications. We

acknowledge that the transition from simulation to physical prototypes requires experimental validation, which is beyond the scope of the current work. The optimized designs are promising candidates for fabrication, but their real-world performance may be affected by factors not captured by our idealized model, such as particle shape imperfections, surface roughness, and environmental conditions. Future work will focus on experimental validation and the integration of additional physical effects (friction, rotation, temperature dependence) into the differentiable simulator to bridge the simulation-to-physical gap.

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